

Math 118– Lecture 02

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1 Norms, Inner Products, Cauchy, etc

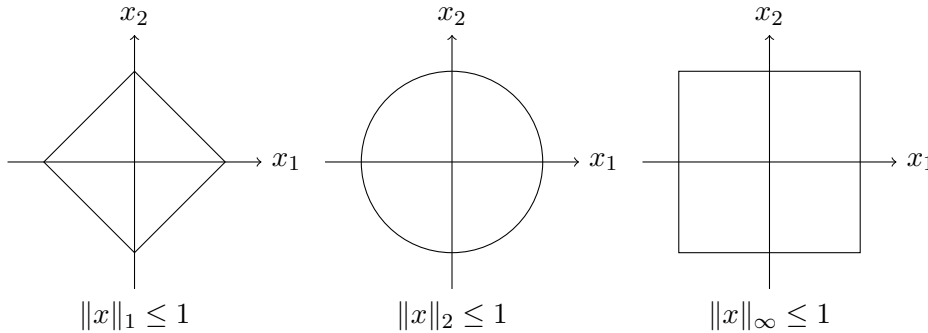
Definition 1.1 (norm). A norm on a vector space V is a mapping $\|\cdot\| : V \rightarrow [0, \infty)$. with properties

- Triangle Inequality: $\|u + v\| \leq \|u\| + \|v\|$.
- Homogeneity $\|\alpha u\| = |\alpha| \|u\|$
- Positive Definiteness. $\|u\| = 0 \iff u = 0$
- Distance from u and v is $\|u - v\|$

1.1 Norms on R^n or C^n

- One-norm: $\|x\|_1 = \sum_{k=1}^n |x_k|$
- Two-norm: $\|x\|_2 = \sqrt{\sum_{k=1}^n (x_k)^2}$
- Max-norm: $\|x\|_\infty = \max_{k \leq n} |x_k|$

Definition 1.2 (Unit Ball). $B = \{x \in V : \|x\| \leq 1\}$



For any $x \in \mathbb{C}^n$, we have

$$\|x\|_\infty \leq \|x\|_2 \leq \|x\|_1$$

because

$$\|x\|_\infty = \sqrt{\max_k |x_k|^2} \leq \sqrt{\sum_k |x_k|^2}$$

$$x = \begin{pmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{pmatrix}, \quad e_j = \begin{pmatrix} 0 \\ \vdots \\ 0 \\ 1 \\ 0 \\ \vdots \\ 0 \end{pmatrix} \quad (1 \text{ in the } j\text{th slot})$$

$$\text{where } (e_j)_i = S_{ij} = \begin{cases} 1, & i = j, \\ 0, & i \neq j. \end{cases}$$

So we can rewrite

$$\|x\|_2 = \|x_1 e_1 + x_2 e_2 + \cdots + x_n e_n\|_2$$

By the triangle inequality we have

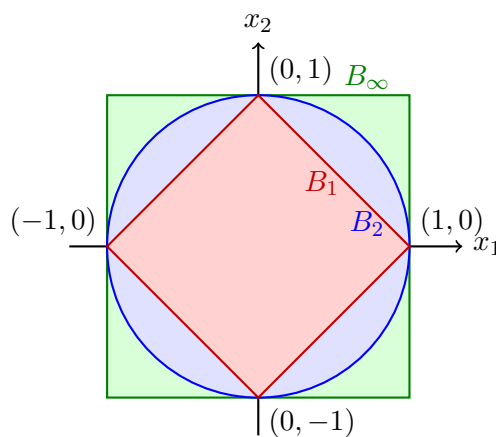
$$\|x_1 e_1 + x_2 e_2 + \cdots + x_n e_n\|_2 \leq \|x_1 e_1\|_2 + \|x_2 e_2\|_2 + \cdots + \|x_n e_n\|_2$$

$$\|x_1 e_1\|_2 + \cdots + \|x_n e_n\|_2 = |x_1| \|e_1\|_2 + |x_2| \|e_2\|_2 + \cdots + |x_n| \|e_n\|_2$$

Notice by definition of e_i its two norm would be 1, so this is just $\sum_i |x_k| = \|x\|_1$

Remark 1.3 (Unit Balls Nested in the opposite direction). We have

$$B_1 \subseteq B_2 \subseteq B_\infty$$



if $x \in B$, then $\|x\|_1 \leq 1$ so

$$\|x\|_2 \leq \|x\|_1 \leq 1$$

2 Inner Products

An inner product is a mapping $\langle \cdot, \cdot \rangle : V \times V \rightarrow \mathbb{C}$ such that

1. $\langle u, u \rangle > 0$ if $u \neq 0$
2. $\langle v, u \rangle = \overline{\langle u, v \rangle}$
3. $\langle \alpha u, v \rangle = \alpha \langle u, v \rangle$.
4. $\langle u + v, w \rangle = \langle u, w \rangle + \langle v, w \rangle$
5. 3 implies that $\langle 0, v \rangle = 0$
6. 2 and 3 imply that $\langle u, \alpha v \rangle = \overline{\alpha} \langle u, v \rangle$
7. 2 and 4 imply that $\langle w, u + v \rangle = \langle w, u \rangle + \langle w, v \rangle$

Geometrically:

$$\langle u, v \rangle = \|u\| \cdot \|v\| \cdot \cos \theta$$

$$\|\alpha u\| = \sqrt{\langle \alpha u, \alpha u \rangle} = \sqrt{\alpha \overline{\alpha} \langle u, u \rangle} = |\alpha| \sqrt{\langle u, u \rangle}$$

In physics, usually the inner product is defined to be linear in y .

Theorem 2.1 (Cauchy Schwartz) *Prove that*

$$|\langle u, v \rangle| \leq \|u\| \cdot \|v\|$$

Show Cauchy, Pythagoras, Corollaries

Definition 2.2 (Orthogonality). Vectors $e_1, e_2, \dots, e_n$ are orthogonal if $\langle e_j, e_k \rangle = S_{jk} = \begin{cases} 1, & j = k, \\ 0, & j \neq k. \end{cases}$

Recall that for an orthonormal basis, $e_1, e_2, \dots, e_n$ is that you can compute the coefficients with an inner product

$$x = \sum_{j=1}^n \alpha_j e_j = \langle \sum_j \alpha_j e_j, e_k \rangle = \sum_j \alpha_j \langle e_j, e_k \rangle$$

You can do Gram - Schmidt orthogonalization to create a new orthonormal basis given linearly independent vectors, and the orthogonal vectors will have the same span.

3 Definitions and notation

Definition 3.1 (First definition). Write the definition and one sentence explaining why it matters.

4 Claims, theorems, and proof sketches

Claim. State the claim in one precise sentence.

Proof sketch. Record the key idea, the first useful equation, and the step that still needs checking.

5 Examples and calculations

Example 5.1 (Lecture example). Write the setup, the calculation, and the conclusion.

6 Questions and follow-up

- What is still unclear?
- Which exercise or reading should I revisit?
- TODO: